

BAAZ & UDAY JKBOSE

2026 CLASS 11TH



ANIMAL KINGDOM

LECTURE-1

ZOOLOGY

DAVOOD SIR (CEO of NG)



Physics Wallah

Topics To Be Covered

1

Basis of Classification

NEET → 12-marks.

Board → 5-marks.

2

Previous Year NEET MCQ's



Basis of Classification

Despite the differences in structure and appearance among animals, there are some fundamental features that are common to many of them. These features form the basis for classifying animals, and include:

① *Level of Organisation.*

Body symmetry.

Body Plan.

Number of Germ Layers.

Nature of coelom (body cavity)

Notochord.

Segmentation

Digestive system

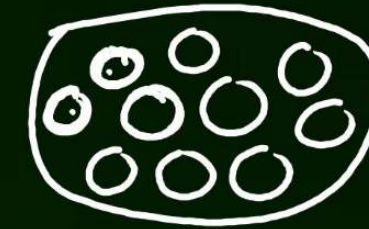
Circulatory system

These basic characteristics help in grouping animals into different phyla.

1. Levels of Organisation

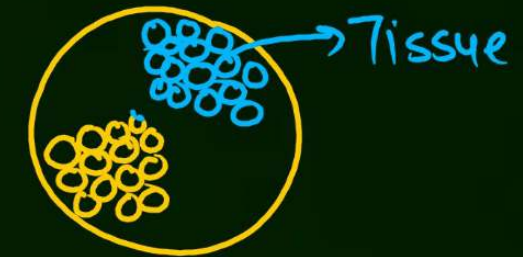
(a) Cellular Level of Organisation

- Body is made of loose aggregate of cells.
- No coordination is present b/w cells.
- Example: Sponges



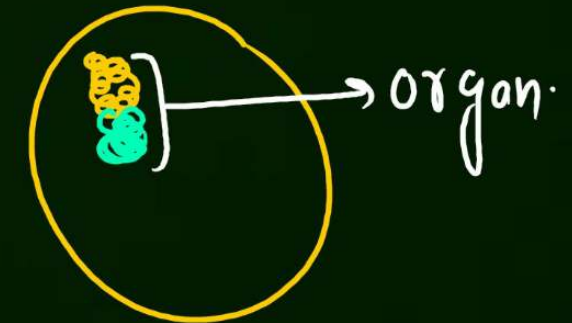
(b) Tissue Level of Organisation:

- Cells performing same function are organised into tissue
- More complex than cellular level
- Example: → Coelenterata (Hydra)



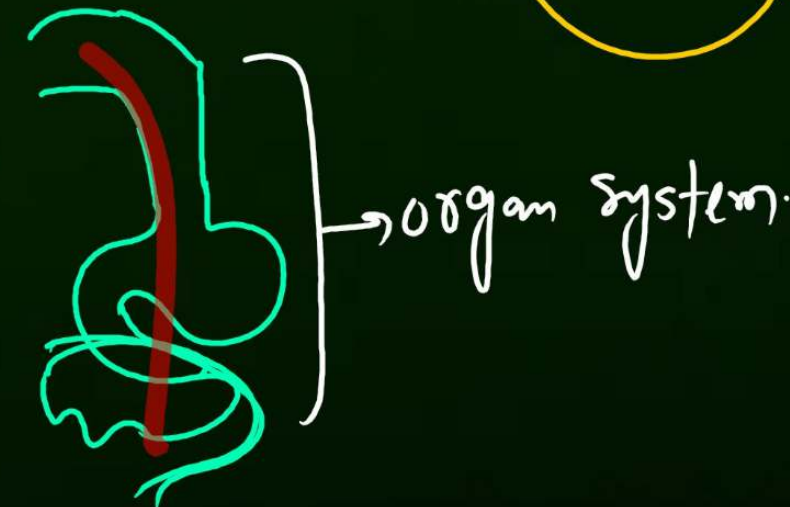
(c) Organ Level of Organisation

- Tissue groups together form organs
- Each organ performs separate function.
- Example: → Platyhelminthes.



(d) Organ System Level of Organisation

- Many organs work together to form organ system & organ system performs unique function.
- Example: → Annelida, Chordata, Mollusca.



2. Symmetry

↳ It is a way by which an animal's body can be divided into two equal halves / Identical halves.

1. Asymmetry Animals



- No plane divides the body into two equal halves because their shape is irregular.
- e.g. Phylum porifera

2. Radial Symmetry



- Any plane passing the central axis divides the body into two identical halves.
- e.g. Echinodermata.
Coelenterata.

3. Bilateral Symmetry



- Body can be divided into left & right halves in only one plane
- e.g. Annelida, Arthropoda

3. Germ Layer Organisation

→ We have divided animals into triploblastic and Diploblastic animals based on the number of germ layers formed during embryonic stage

1. Diploblastic Animals

2

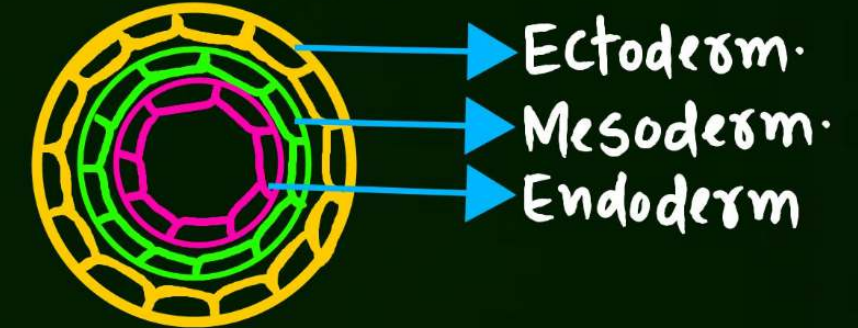


- Body is made of two germ layers i.e Ectoderm and endoderm. Mesoderm is absent instead of it a non-cellular layer Mesoglia is present in b/w these two.

Example: → Coelenterata (Hydra).

2. Triploblastic Animals

3



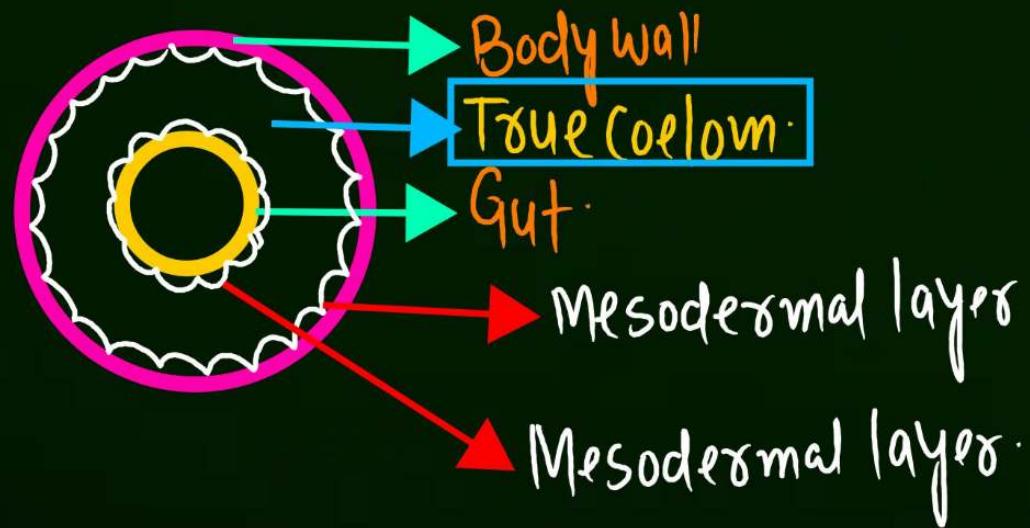
- Body is made of three germ layers i.e Ectoderm, Mesoderm and endoderm.
Example: → Platyhelminthes upto chordates

4. Coelom

- It is the fluid filled cavity present b/w body wall and gut.
- it is lined Mesoderm by both side

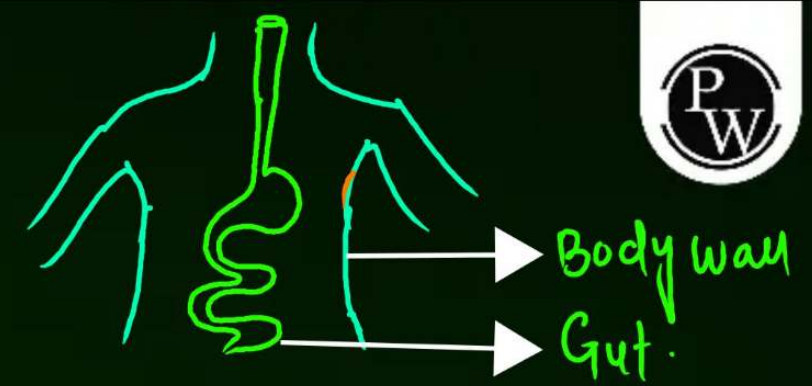
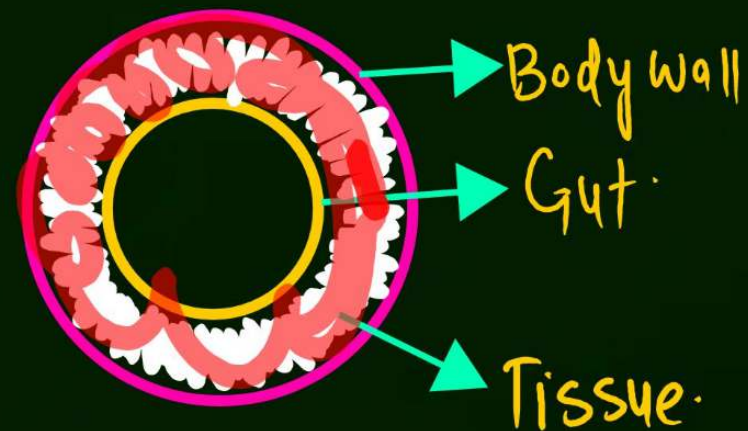
1. Coelomates.

- True Coelom is present and is completely lined by mesoderm.
- E.g.: → Annelida, Chordata.



2. Acoelomates:

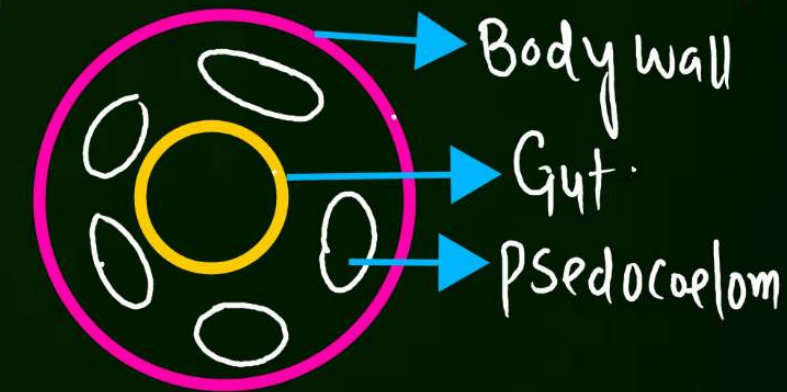
- No body cavity of Coelom is present because the Coelom is filled with tissue.
- E.g.: → Platyhelminthes (Flatworms).



3. Pseudocoelomates

False

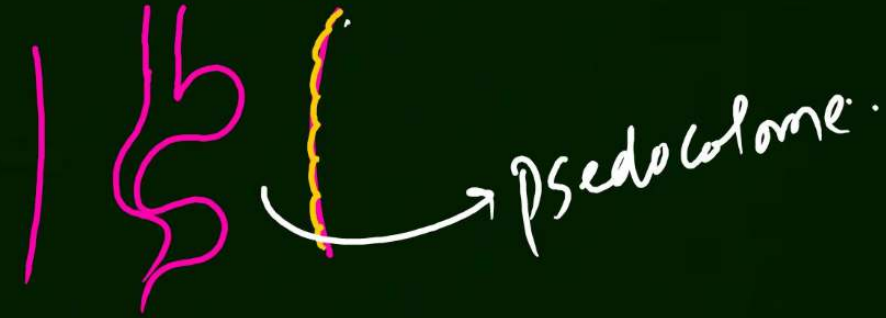
- Body cavity is not full lined by mesoderm and forms pouches b/w ectoderm and endoderm.
- E.g. Aschelminthes (Roundworms).



Question [NEET-2025]

Kaun Banayga Nabkaar

While trying to find out the characteristic of a newly found animal, a researcher did the histology of adult animal and observed a cavity with presence of mesodermal tissue towards the body wall but no mesodermal tissue was observed towards the alimentary canal. What could be the possible coelome of that animal?



1 Pseudocoelomate ✓

2 Schizocoelomate ✗

3 Spongocoelomate ✗

4 Acoelomate ✗

Kaun Banayga Nabkaar : NEET-2023



Radial symmetry is not found in:



A

Coelentrata

B

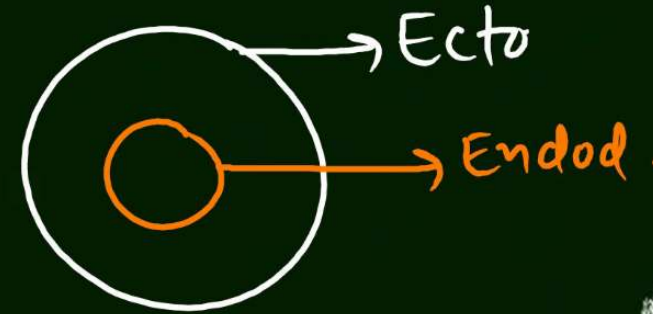
Echinodermata

C

Ctenophora

D

Hemichordata. ✓



Thank
You